Examining the bidirectional relationship between sleep and physical activity using pooled individual data

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Summary

Background Physical activity is recommended to improve sleep in adults, but evidence in children and adolescents is unclear and largely self-reported. We examined day-to-day, bidirectional associations between device-measured physical activity and sleep across the lifespan.

Methods We pooled individual participant data from studies using research-grade accelerometers worn 24 hours a day for at least four days. We derived daily physical activity volume and intensity, and sleep duration, efficiency, onset, and regularity, from harmonised raw data. Mixed-effects models with age interactions separated within-person (day-to-day) from between-person (habitual) associations of physical activity with that night's sleep, and of sleep with next-day activity, adjusting for sex, BMI, and socioeconomic status. The protocol was pre-registered (https://osf.io/ahd96).

Findings In 27,236 participants aged 2 to 94 years from 20 studies, participants slept longer and fell asleep earlier on days when they were more active than usual. A 1 SD higher volume than usual was associated with 9.6 min longer sleep (standardised β [95% CI] 0.11 [0.10, 0.12]) and 12 min earlier sleep onset at age 11 years, and with 8.6 min and 20 min, respectively, at age 65 years. A night's sleep was barely associated with next-day activity (all $|\beta| < 0.03$). Habitually more active participants had more regular sleep, but the day-level association with sleep regularity reflected how the regularity index was measured.

Interpretation At every age, people slept slightly longer and earlier after more active days. Physical activity could be promoted to support sleep in children and adolescents as well as adults, as a small, consistent benefit rather than a remedy for poor sleep.

Funding This research was unfunded.

Research in context

Evidence before this study

Meta-analyses of trials in adults show that physical activity improves sleep, with small effects on sleep duration and efficiency and the largest benefits in people who sleep poorly, and the WHO recommends physical activity for this purpose. In children and adolescents the evidence is less clear. Meta-analyses of observational studies find inconsistent associations. Trials suggest that exercise improves sleep quality and efficiency but not total sleep time, and guidelines do not recommend physical activity to improve sleep in these age groups. Most studies relied on self-reported sleep or physical activity, and meta-analyses that pool study-level effects cannot show how associations change with age. Whether the relationship runs in both directions, and whether people sleep better after more active days rather than only being more active in general, needs device-measured, day-level data. These have been synthesised separately in adults, where a meta-analysis of 33 studies did not support a bidirectional daily association, and in children and youth, where a meta-analysis of 11 studies found weak and inconsistent associations. Recent studies of consumer wearables with millions of person-nights included only adults and did not examine sleep regularity.

Added value of this study

To our knowledge, this is the first study to examine day-to-day associations between device-measured physical activity and sleep in a large, geographically diverse sample spanning the lifespan. We reprocessed raw accelerometer data from 20 studies with a common method, giving daily data for 27,236 participants aged 2 to 94 years. By separating day-to-day from habitual differences, we found that on days when people were more active than usual they slept longer and fell asleep earlier that night, from childhood to older age. These associations were small, under 10 minutes of sleep per 1 SD of activity volume, and a night's sleep was barely associated with next-day activity. Habitually more active people had more regular sleep.

Implications of all the available evidence

Trials show that physical activity improves sleep in adults, and our findings show that people of all ages, including children and adolescents, sleep slightly longer and earlier after more active days. Public health messages could present physical activity as a small, consistent support for sleep at every age, rather than as a remedy for poor sleep. That the day-level associations varied little with age suggests that the weaker evidence in children and adolescents may reflect study design or measurement rather than age. Trials should test whether increasing physical activity lengthens sleep in children and adolescents, and studies of sleep regularity should use measures that separate sleep from daytime rest.

Introduction

Sleep is an important determinant of population health across the lifespan: poor sleep is associated with all-cause, cardiovascular disease, and cancer mortality[1] and with poor mental health.[2] Yet 14-68% of children and adolescents[3] and 31% of adults[4] sleep less than recommended, and safe, accessible strategies to promote sleep are needed.[5,6] In adults, physical activity appears to improve both the quantity and quality of sleep,[7,8] and WHO recommends it for this purpose.[9] In children and adolescents, the evidence is less clear.[6,10,11] Trials suggest exercise improves sleep quality and efficiency but not total sleep time,[12] and guidelines do not support a role for physical activity in improving sleep in this age group.

The existing evidence has three main limitations. First, most studies relied on self-reported sleep and physical activity,[6,10,11] which could bias associations and limit the precision of dose-response analyses. Second, existing reviews and meta-analyses[6] pool study-level effects, which limits how well they can examine the influence of age.[13] Third, no study has examined day-to-day data in a large, geographically diverse sample spanning the lifespan. Daily associations have been synthesised separately in adults[14] and in children and youth,[15] and consumer-wearable studies with millions of person-nights included only adults and did not examine sleep regularity.[16,17] Day-level data matter for two reasons. First, they separate whether people who are habitually more active sleep better (a between-person association, open to confounding by stable characteristics) from whether people sleep better after days on which they are more active than usual (a within-person association, closer to the effect of changing one's activity). Second, the relationship may run in both directions: physical activity may promote sleep through circadian and stress-related pathways,[18] while a late or poor night's sleep may reduce activity the next day.[14] Pooling device-measured individual participant data can overcome these limitations.[13]

We harmonised individual participant data from accelerometer studies around the world to investigate the relationship between device-measured physical activity and sleep across the lifespan. We addressed three research questions (RQs): (RQ1) do individuals who are more physically active have better sleep duration and quality; (RQ2) does age moderate the association between physical activity and sleep; and (RQ3) is the association bidirectional?

Methods

We pre-registered all methods and analyses on the Open Science Framework (OSF) before data acquisition began (https://osf.io/ahd96).

Data sources and data acquisition

We included studies of any design that used research-grade accelerometers of any brand, worn at any location (hip, wrist, or thigh), from which raw accelerations (m/s^2 or mg) could be obtained, with a 24-hour wear protocol over at least four days, including one weekend day.

We identified potentially relevant studies by examining the reference lists of recent systematic reviews that explored the relationship between sleep and physical activity in children, adolescents, and adults.[7,10,11] We then used forward searching on potentially eligible studies. We also searched for accelerometry consortiums, searched trial registries (ANZCTR, WHO ICTRP, and ISRCTN) and the Open Science Framework for unpublished results, and asked collaborators to identify additional studies.

To keep data acquisition manageable, we invited only studies with more than 400 participants, emailing chief investigators or consortium leads to request deidentified individual data. All studies included in the pooled data meta-analysis had to have been approved by their local or institutional ethical committees, and all participants (or, for children, their parents or guardians) had to have provided written informed consent, with children providing assent where required. The Australian Catholic University's Research Ethics Committee granted permission for each study's data to be obtained (approval: 2020-143N), and investigators transferred their deidentified data via a secure file-sharing system subject to original institutional ethics approvals or new data transfer agreements.

Data harmonisation and outcomes

We harmonised all accelerometer data by reprocessing raw files with the open-source *GGIR* package (version 2.5-1) in R,[19,20] to produce comparable outcomes across studies, device wear locations, and populations. Processing used our *sleepIPD* R package, which fixed settings across studies and documented the decisions made for specific devices.[21] The core research team or the data contributors ran *sleepIPD* on each study's raw files. *GGIR*'s sleep detection has been evaluated against polysomnography in both adults[20] and children.[22] Non-wear time and abnormally high accelerations were detected by default settings, classified as missing, and imputed from the same time interval on the remaining days of the week.[19]

Each day ran from waking to waking the following day, so each day's physical activity, calculated over its waking hours, was paired with the sleep period that followed it. For wrist-worn devices, the sleep period was detected with the HDCZA algorithm[20] and for hip-worn devices *GGIR* instead detected time in bed from the device's angle. For sleep, *GGIR* produced total sleep time (minutes), sleep efficiency (percentage of the sleep period, from sleep onset to final waking, spent asleep), sleep onset (clock time), and sleep regularity (the sleep regularity index: the likelihood of being in the same sleep or wake state at time points 24 hours apart, with higher values indicating more regular sleep). *GGIR* calculates the index by comparing each calendar day with the next and, in the version we used, counts any sustained inactivity as sleep, including daytime rest.[23] For physical activity, we used two metrics proposed by Rowlands and colleagues:[24] average acceleration (mg)—a measure of volume—and the intensity gradient (unitless), the slope of log time against log intensity, which describes how time is distributed across intensities, with less negative values indicating relatively more time at higher intensities. We chose these metrics because cut-points for moderate-to-vigorous physical activity (MVPA) depend on age, device, and wear location; those applied to our data differed more than two-fold across groups (appendix), so minutes of MVPA are not directly comparable across studies. Average acceleration and the intensity gradient need no cut-points, capture light as well as vigorous activity, and separate how much activity was done from how it was distributed across intensities. To aid interpretation, we also analysed daily minutes of MVPA, from *GGIR*'s age-, device-, and location-specific cut-points,[21] as a secondary exposure.

Covariates and moderators

The following individual participant data were obtained when available: age, sex, height, weight, waist circumference, maturational status, socioeconomic status (SES), ethnicity, screen time, sleep medications, and medically diagnosed conditions known to affect sleep. Sex was ascertained from the original studies, which primarily used self- or parent-report for sex assigned at birth. The country, city, time zone, daylight length, and season in which data were collected

were also obtained from the original studies. Because studies measured socioeconomic status differently (e.g., education or income), we harmonised each study's measure into low, medium, and high categories (see appendix).

Body mass index (BMI) was converted to an age- and sex-standardised z-score. For participants under 18 years we used the CDC 2000 BMI-for-age growth reference, applied with the *childsds* package. For adults we standardised BMI against the adults in this sample.

Statistical analysis

To reduce the risk of bias, we used a blind analysis approach: we developed and tested models on a 10% development subsample of the participants; we conducted the final analysis once we determined the code was ready. Data screening removed values outside fixed physiological limits (for average acceleration, above 200 mg), all physical activity data from files indicating device failure (76 participants), values more than 4 SD from their own study's mean, and biologically implausible growth values in participants under 18 years (see appendix).

A day was valid if it recorded more than 10 hours of wear time and, where sleep was detected, a sleep period longer than 200 minutes. Participants required at least 4 valid days. A sensitivity analysis restricted the sample to the 14,981 participants with at least seven valid days.

We assumed missing covariate and moderator data to be missing at random or completely at random and imputed them using multilevel predictive mean matching, clustered by participant, with multiple imputation by chained equations in the *mice* package, creating 50 imputed datasets. The imputation model also included the sleep and physical activity variables, but each analysis used only the person-days on which its outcome, its sleep or physical activity predictor, and its moderator were observed, so imputed values are only covariates. We pooled results across imputed datasets with Rubin's rules, computing 95% CIs and p values on the Barnard-Rubin degrees of freedom, and used $p<0.05$ to assess statistical significance. We fitted 16 models for each moderator and report all of them without adjustment for multiple comparisons, so p values close to 0.05 should be interpreted with caution. We used R (version 4.6.1) for all data cleaning and analysis.

To examine the relationship between sleep and physical activity we ran mixed-effects models with random intercepts for study and participant, accounting for the nesting of days within participants and of participants within studies. We split each predictor into the participant's mean across their valid days (the between-person component) and that day's deviation from the mean (the within-person component), and entered both, so that within-person associations were not confounded by stable differences between people. We predicted each sleep indicator from each physical activity indicator, adjusting for SES, sex, and BMI z-score, with quadratic terms to allow curvilinear associations. We interacted each component with age to examine how associations change over the lifespan (RQ2). Age was centred at its mean across person-days (35.6 years) and scaled per 10 years. For interpretability, we report associations at ages 11, 18, 35, and 65 years. Models predicting physical activity from the preceding night's sleep (RQ3) used each sleep indicator from the previous day, so person-days whose preceding day was absent or ineligible did not contribute, leaving 136,319 of 168,613 eligible person-days. Because the previous day's sleep regularity index also spans the day being predicted, we instead used the index for the two preceding calendar days. Sleep regularity was missing most often, so its models with physical activity volume used 125,155 person-days (RQ1) and 86,677 (RQ3). We also ran

a cross-lagged sensitivity analysis, adjusting each model for the outcome's previous-day value, and tested whether next-day activity, which cannot affect the night's sleep, predicted each sleep outcome (i.e., negative control; see appendix).

Protocol deviations

This paper addresses the first two hypotheses in our registered protocol: that physical activity volume and intensity would be positively associated with optimum sleep duration and quality, with an inverted U-shaped dose-response, and that associations would be stronger in adolescents and adults than in children and older adults. Moderation by the other pre-registered moderators is reported in the appendix, and the remaining hypotheses will be addressed in future research. While testing with the development subsample we made three changes to the protocol. We modelled study as a random rather than a fixed effect, so that estimates would generalise beyond the included studies. Sensitivity analyses of fixed-effect models gave qualitatively similar results. We analysed each sleep metric separately, because no validated composite sleep indicator could be estimated from our data. We did not log-transform skewed variables, because back-transformed predictions followed extreme and implausibleexponential patterns near the edges of the data range. Log-scale models are reported in the appendix. Additionally, after running the full analysis we identified a mistake in the BMI variable and in the imputation model, which we corrected and reran the analysis. After data contributor review of the initial results, we separated within- and between-person associations, which the protocol did not specify, and realigned the sleep regularity index in RQ3 models. The pre-registered models, which combine within- and between-person associations, are reported in the appendix.

Role of the funding source

None.

Results

The pooled dataset contained 220,988 days of physical activity and sleep data from 28,824 participants in 20 studies. We excluded 52,375 days with insufficient wear time, which removed 1,175 participants entirely, and a further 413 participants (619 days) with fewer than 4 valid days, leaving 27,236 participants and 167,994 person-days (Table 1; study characteristics are in the appendix). Participants were aged 2 to 94 years and 50.3% were female. Most came from Europe and South America, followed by Oceania, and the sample skewed towards younger children (2-11 years, 50.9%) and older adults (66+ years, 22.3%).

Table 1: Participant characteristics

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	13,820	1,923	310	5,044	6,067
Age	8.5 (2.8)	15.7 (1.7)	29.2 (5.1)	57.7 (7.4)	72.7 (4.5)
BMI	17.7 (3.0)	21.6 (3.8)	26.5 (6.5)	26.8 (4.8)	27.1 (4.3)
BMI z-score	0.31 (1.10)	0.09 (1.02)	−0.07 (1.40)	−0.01 (1.03)	0.05 (0.93)
Valid Weartime Hours	23.4 (1.3)	23.4 (1.5)	23.3 (1.7)	23.8 (0.7)	23.9 (0.4)
Valid Weartime Days	5.7 (1.8)	5.0 (1.3)	5.0 (1.4)	7.5 (2.3)	6.6 (2.1)
PA Intensity Gradient (unitless)	−2.10 (0.18)	−2.23 (0.19)	−2.43 (0.19)	−2.50 (0.19)	−2.68 (0.22)
PA Volume (average acceleration in mg)	49.0 (26.4)	46.7 (14.3)	39.3 (10.6)	38.8 (11.6)	30.9 (9.7)
Sleep Duration (min)	471.7 (61.5)	416.4 (55.0)	421.0 (64.0)	409.1 (55.8)	404.3 (63.9)
Sleep Efficiency (%)	80.9 (7.0)	87.1 (6.3)	87.4 (6.2)	87.8 (5.5)	86.4 (6.2)
Sleep Onset (clock time; SD in h:mm)	21:32 (01:42)	23:29 (01:27)	23:33 (01:49)	23:25 (01:17)	23:57 (01:16)
Sleep Regularity Index	53.4 (12.2)	57.0 (12.1)	55.8 (12.8)	59.4 (11.4)	54.5 (12.5)
Accelerometer Wear Location					
Wrist	5,509 (39.9%)	1,885 (98.0%)	310 (100.0%)	5,044 (100.0%)	6,067 (100.0%)

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Hip	8,311 (60.1%)	38 (2.0%)	-	-	-
Country					
Australia	1,510 (10.9%)	305 (15.9%)	110 (35.5%)	188 (3.7%)	-
Finland	508 (3.7%)	-	-	801 (15.9%)	-
New Zealand	455 (3.3%)	-	37 (11.9%)	152 (3.0%)	-
Spain	809 (5.9%)	-	-	37 (0.7%)	2,386 (39.3%)
Switzerland	-	-	-	1,905 (37.8%)	1,023 (16.9%)
United Kingdom	447 (3.2%)	-	144 (46.5%)	1,961 (38.9%)	2,658 (43.8%)
Brazil	3,425 (24.8%)	727 (37.8%)	10 (3.2%)	-	-
Czech Republic	281 (2.0%)	848 (44.1%)	9 (2.9%)	-	-
Canada	522 (3.8%)	-	-	-	-
Chile	138 (1.0%)	5 (0.3%)	-	-	-
China	125 (0.9%)	-	-	-	-
Colombia	889 (6.4%)	-	-	-	-
Denmark	689 (5.0%)	-	-	-	-
Ecuador	981 (7.1%)	38 (2.0%)	-	-	-
India	575 (4.2%)	-	-	-	-
Kenya	528 (3.8%)	-	-	-	-
Portugal	702 (5.1%)	-	-	-	-

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
South Africa	507 (3.7%)	-	-	-	-
United States	729 (5.3%)	-	-	-	-
Ethnicity					
Non-White	2,672 (19.3%)	169 (8.8%)	14 (4.5%)	94 (1.9%)	240 (4.0%)
Other or unknown	4,780 (34.6%)	1,194 (62.1%)	154 (49.7%)	1,340 (26.6%)	2,424 (40.0%)
White	6,368 (46.1%)	560 (29.1%)	142 (45.8%)	3,610 (71.6%)	3,403 (56.1%)
Region					
Europe	3,436 (24.9%)	848 (44.1%)	153 (49.4%)	4,704 (93.3%)	6,067 (100.0%)
Oceania	1,965 (14.2%)	305 (15.9%)	147 (47.4%)	340 (6.7%)	-
South America	5,433 (39.3%)	770 (40.0%)	10 (3.2%)	-	-
Africa	1,035 (7.5%)	-	-	-	-
Asia	700 (5.1%)	-	-	-	-
North America	1,251 (9.1%)	-	-	-	-
Season					
Autumn	4,672 (33.8%)	445 (23.1%)	39 (12.6%)	1,241 (24.6%)	1,268 (20.9%)
Spring	2,859 (20.7%)	1,230 (64.0%)	125 (40.3%)	1,226 (24.3%)	1,988 (32.8%)
Summer	2,113 (15.3%)	82 (4.3%)	85 (27.4%)	1,271 (25.2%)	1,252 (20.6%)
Winter	4,176 (30.2%)	166 (8.6%)	61 (19.7%)	1,306 (25.9%)	1,559 (25.7%)
Sex					

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Female	7,154 (51.8%)	1,002 (52.1%)	222 (71.6%)	2,717 (53.9%)	2,551 (42.0%)
Male	6,664 (48.2%)	921 (47.9%)	88 (28.4%)	2,327 (46.1%)	3,516 (58.0%)
Sleep Conditions Reported					
No	625 (4.5%)	-	-	3,512 (69.6%)	5,141 (84.7%)
Yes	50 (0.4%)	-	-	310 (6.1%)	626 (10.3%)
Not Reported	13,145 (95.1%)	1,923 (100.0%)	310 (100.0%)	1,222 (24.2%)	300 (4.9%)
Socioeconomic Status					
Low	2,892 (24.1%)	208 (11.5%)	2 (0.7%)	1,074 (22.0%)	2,388 (39.4%)
Medium	4,638 (38.7%)	885 (49.1%)	191 (70.2%)	2,409 (49.3%)	2,325 (38.3%)
High	4,456 (37.2%)	708 (39.3%)	79 (29.0%)	1,406 (28.8%)	1,352 (22.3%)

For categorical variables the value is the count, and percentage. For numeric variables the value is the Mean and SD. Sleep onset is a clock time; its SD is a duration in hours:minutes. Total N = 27,236. 72 participants had missing age data and are counted in the total but not in any age band.

Physical activity and that night's sleep

On days when participants were more active than usual, they slept longer and fell asleep earlier that night (Table 2; Figure 1). These within-person associations changed little with age for sleep duration (β = 0.11 [0.10, 0.12] at age 11 and 0.10 [0.08, 0.12] at age 65) and strengthened with age for sleep onset (β = −0.10 [−0.11, −0.09] and −0.16 [−0.17, −0.15]). Within-person associations with sleep efficiency were small, positive at age 11 (β = 0.03 [0.03, 0.04]) and negative at age 65 (β = −0.06 [−0.07, −0.04]). Between people, habitually more active participants slept longer at age 11 (β = 0.17 [0.14, 0.19]) but slightly less at age 65 (β = −0.05 [−0.08, −0.02]), so the weakening with age of the association with sleep duration in the pre-registered models (appendix) reflected differences between people. Curvilinear terms were statistically significant but did not meaningfully change these associations.

Activity volume was also associated with sleep regularity, within (β = 0.18 [0.17, 0.19] at age 11 and 0.32 [0.29, 0.34] at age 65) and between people (β = 0.41 [0.39, 0.44] and 0.48 [0.43, 0.52]). However, in the negative control, next-day activity was associated with the night's sleep regularity index at least as strongly as same-day activity (β = 0.23 [0.22, 0.25] *vs* 0.19 [0.18, 0.20]), whereas next-day associations with sleep duration (0.03 [0.03, 0.04] *vs* 0.10 [0.09, 0.11]) and onset (-0.02 [−0.03, −0.02] *vs* −0.10 [−0.11, −0.09]) were much smaller (appendix). Days of unusually low and unusually high activity were both associated with lower regularity (Figure 1), so the within-person association with the index cannot be attributed to sleep.

Table 2: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.04 [−0.20, 0.11]	.592	−0.04 [−0.17, 0.10]	.609
Physical activity (within)	0.10 [0.09, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Age	−0.04 [−0.06, −0.03]	< .001	−0.03 [−0.04, −0.01]	< .001
Physical activity (within)²	0.01 [0.00, 0.03]	.113	0.00 [−0.01, 0.01]	.908
Physical activity (between)	0.07 [0.05, 0.09]	< .001	0.11 [0.09, 0.13]	< .001
Physical activity (between)²	0.00 [−0.01, 0.02]	.883	−0.02 [−0.03, 0.00]	.024
Physical activity (within) × Age	0.00 [−0.01, 0.00]	.260	−0.01 [−0.01, 0.00]	< .001
Age × Physical activity (within)²	0.01 [0.00, 0.01]	.002	0.00 [−0.01, 0.00]	.012
Age × Physical activity (between)	−0.04 [−0.05, −0.03]	< .001	−0.03 [−0.04, −0.02]	< .001
Age × Physical activity (between)²	0.01 [0.00, 0.01]	.003	0.00 [0.00, 0.00]	.771
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.34]	.017	0.17 [0.00, 0.35]	.046
Physical activity (within)	−0.01 [−0.02, 0.00]	.164	0.00 [−0.01, 0.01]	.987
Age	0.01 [−0.01, 0.02]	.219	0.00 [−0.02, 0.01]	.705

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within)²	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	0.10 [0.08, 0.12]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity (between)²	−0.04 [−0.06, −0.03]	< .001	−0.05 [−0.06, −0.03]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.019	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	.002
Age × Physical activity (between)²	0.00 [0.00, 0.01]	.630	0.01 [0.01, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.13 [−0.14, 0.40]	.359	0.16 [−0.10, 0.42]	.223
Physical activity (within)	−0.13 [−0.13, −0.12]	< .001	−0.09 [−0.09, −0.08]	< .001
Age	0.00 [−0.01, 0.02]	.425	0.01 [0.00, 0.02]	.214
Physical activity (within)²	0.00 [−0.01, 0.01]	.961	0.00 [−0.01, 0.01]	.862
Physical activity (between)	−0.10 [−0.12, −0.09]	< .001	−0.07 [−0.09, −0.05]	< .001
Physical activity (between)²	−0.02 [−0.03, 0.00]	.024	−0.02 [−0.03, 0.00]	.024

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.179
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.005	0.01 [0.01, 0.01]	< .001
Age × Physical activity (between)	0.00 [0.00, 0.01]	.373	0.02 [0.01, 0.03]	< .001
Age × Physical activity (between)²	0.00 [−0.01, 0.00]	.044	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.26 [0.12, 0.39]	< .001	0.08 [−0.11, 0.27]	.404
Physical activity (within)	0.24 [0.23, 0.26]	< .001	0.09 [0.08, 0.09]	< .001
Age	0.04 [0.02, 0.06]	< .001	−0.07 [−0.09, −0.05]	< .001
Physical activity (within)²	−0.31 [−0.33, −0.28]	< .001	−0.13 [−0.14, −0.12]	< .001
Physical activity (between)	0.44 [0.41, 0.47]	< .001	0.15 [0.13, 0.18]	< .001
Physical activity (between)²	−0.19 [−0.21, −0.16]	< .001	−0.13 [−0.15, −0.11]	< .001
Physical activity (within) × Age	0.03 [0.02, 0.03]	< .001	−0.02 [−0.02, −0.02]	< .001
Age × Physical activity (within)²	−0.07 [−0.08, −0.06]	< .001	0.03 [0.03, 0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity (between)	0.01 [0.00, 0.02]	.019	−0.07 [−0.08, −0.06]	< .001
Age × Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001	0.00 [0.00, 0.01]	.105

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

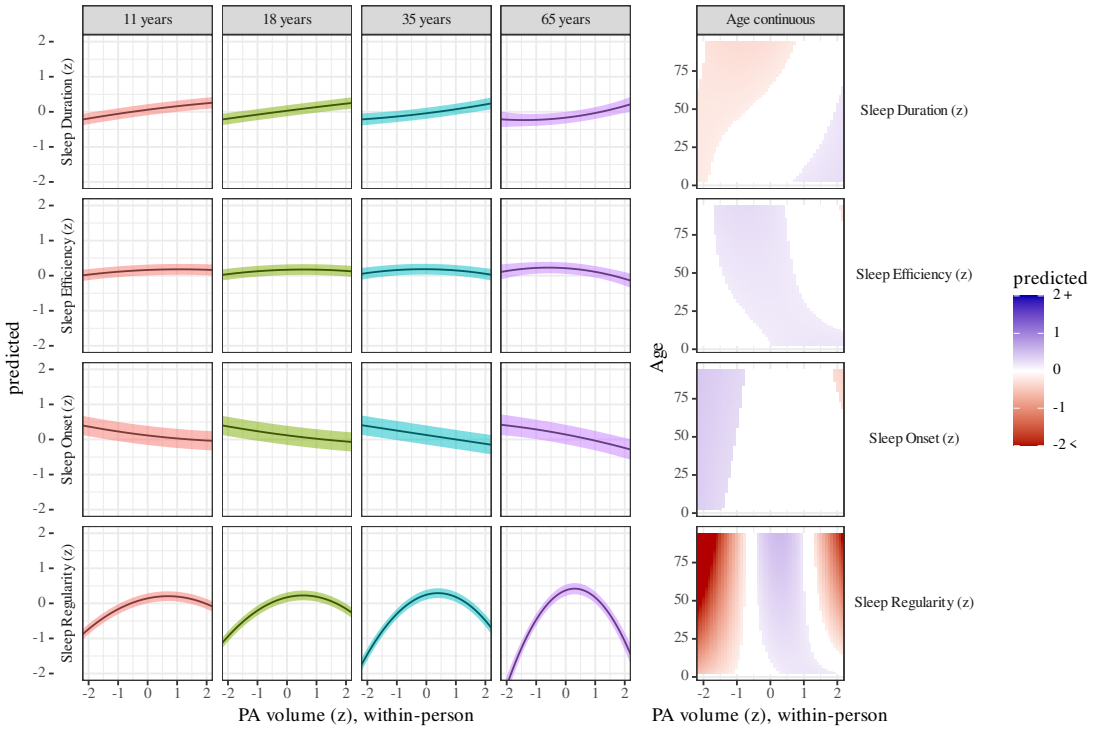


Figure 1: Standardised sleep metrics predicted by within-person physical activity volume (the day's deviation from the participant's mean), with between-person volume held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Within-person associations of physical activity intensity followed the same pattern, with longer sleep (β = 0.08 [0.07, 0.09] at age 11 and 0.04 [0.03, 0.05] at age 65) and earlier sleep onset (β = −0.09 [−0.10, −0.08] and −0.08 [−0.09, −0.08]; Figure 2).

Minutes of MVPA showed a different pattern for sleep timing (appendix). Thirty minutes more MVPA than usual was associated with sleep onset 5 min later at age 11 (β = 0.04 [0.04, 0.05]) and 4 min later at age 65 (β = 0.04 [0.03, 0.04]), and with sleep 1.8 min longer at age 11 but 3.0 min shorter at age 65. Adjusting for the previous day's outcome in cross-lagged models changed the within-person estimates little (appendix).

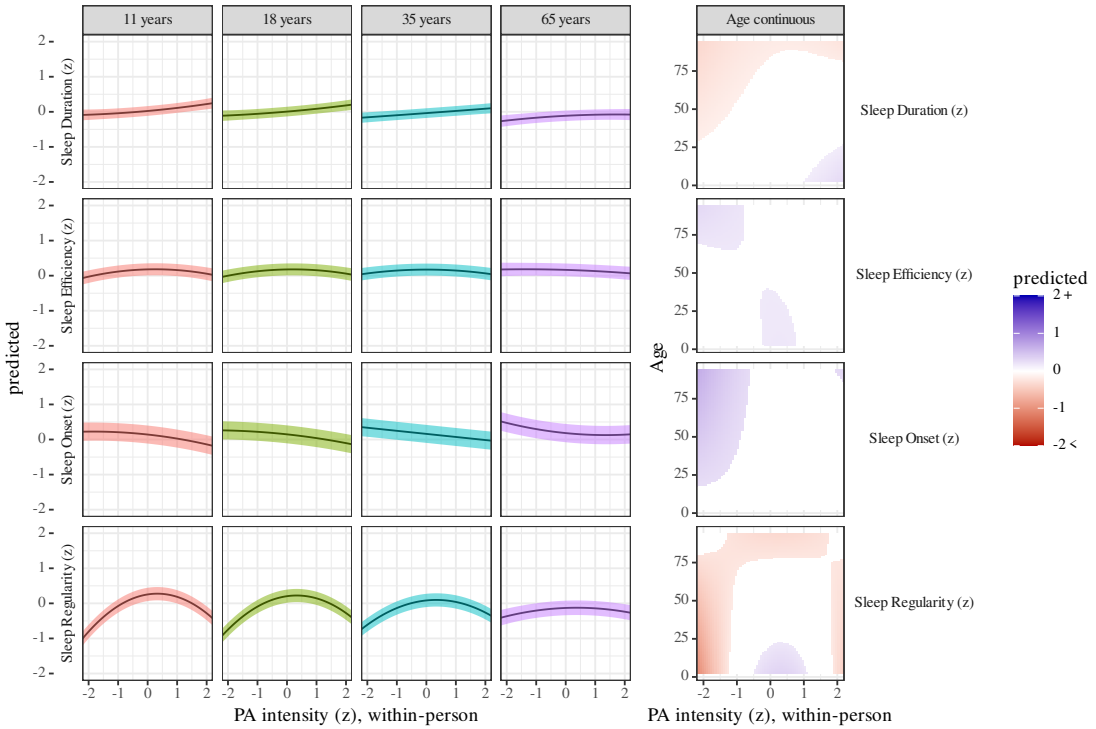


Figure 2: Standardised sleep metrics predicted by within-person physical activity intensity (the day's deviation from the participant's mean), with between-person intensity held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Sleep and next-day physical activity

Within-person associations of a night's sleep with next-day physical activity volume and intensity were negligible at every age (Table 3; Figure 3 to Figure 6). The largest was for sleep duration and next-day volume at age 11 (β = 0.03 [0.02, 0.03]). Sleep regularity over the two preceding days was not associated with next-day volume (β = −0.01 [−0.01, 0.00] at age 11 and −0.01 [−0.02, 0.00] at age 65). Between people, those with habitually more regular sleep were more active (β = 0.16 [0.15, 0.18] at age 11 and 0.18 [0.16, 0.20] at age 65). For MVPA, a 1 SD earlier sleep onset than usual was followed by 3.2 min more MVPA the next day at age 11 and 6.0 min more at age 65, and 1 SD longer sleep by 1.0 min and 3.6 min less, respectively (appendix).

Table 3: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.45]	.379	−0.07 [−0.20, 0.05]	.252
Sleep duration (within, prior)	0.01 [0.01, 0.02]	< .001	0.02 [0.01, 0.02]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration (within, prior)²	−0.04 [−0.05, −0.04]	< .001	−0.02 [−0.03, −0.02]	< .001
Sleep duration (between)	0.07 [0.06, 0.08]	< .001	0.06 [0.05, 0.07]	< .001
Sleep duration (between)²	−0.05 [−0.06, −0.04]	< .001	−0.01 [−0.02, 0.00]	.012
Sleep duration (within, prior) × Age	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.010
Age × Sleep duration (within, prior)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Age × Sleep duration (between)	−0.03 [−0.04, −0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Sleep duration (between)²	0.00 [0.00, 0.00]	.885	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.43]	.392	−0.07 [−0.19, 0.05]	.252

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	.003
Age	−0.23 [−0.24, −0.22]	< .001	−0.28 [−0.29, −0.27]	< .001
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	−0.02 [−0.03, 0.00]	.008	0.01 [0.00, 0.02]	.254
Sleep efficiency (between)²	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.902	0.00 [−0.01, 0.00]	< .001
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.070	0.00 [0.00, 0.00]	.323
Age × Sleep efficiency (between)	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.016
Age × Sleep efficiency (between)²	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.01]	.113
Sleep onset (z)				
(Intercept)	0.13 [−0.20, 0.45]	.444	−0.08 [−0.22, 0.05]	.222
Sleep onset (within, prior)	0.00 [−0.01, 0.00]	.123	0.00 [0.00, 0.01]	.202
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.11 [−0.12, −0.09]	< .001	−0.07 [−0.08, −0.06]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (between)²	−0.01 [−0.01, 0.00]	.153	0.01 [0.00, 0.01]	.105
Sleep onset (within, prior) × Age	0.00 [−0.01, 0.00]	< .001	0.01 [0.00, 0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.001
Age × Sleep onset (between)	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.02]	< .001
Age × Sleep onset (between)²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.02 [−0.27, 0.32]	.874	−0.17 [−0.30, −0.04]	.009
Sleep regularity (within, prior)	−0.01 [−0.01, 0.00]	.067	0.01 [0.00, 0.01]	.053
Age	−0.23 [−0.25, −0.21]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.643	0.00 [−0.01, 0.00]	.251
Sleep regularity (between)	0.17 [0.16, 0.18]	< .001	0.07 [0.06, 0.08]	< .001
Sleep regularity (between)²	0.00 [−0.01, 0.01]	.632	−0.02 [−0.03, −0.01]	< .001
Sleep regularity (within, prior) × Age	0.00 [0.00, 0.00]	.878	0.00 [−0.01, 0.00]	.002

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep regularity (within, prior)²	0.00 [0.00, 0.00]	.209	0.00 [−0.01, 0.00]	< .001
Age × Sleep regularity (between)	0.00 [0.00, 0.01]	.108	0.00 [−0.01, 0.00]	.251
Age × Sleep regularity (between)²	0.00 [0.00, 0.00]	.380	0.00 [−0.01, 0.00]	.003

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

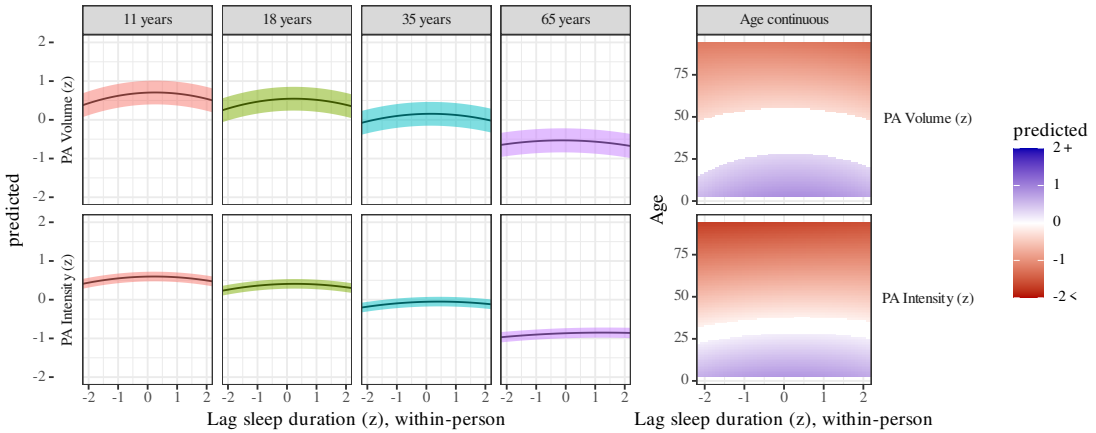


Figure 3: Standardised physical activity predicted by within-person sleep duration the previous night (the deviation from the participant's mean), with between-person sleep duration held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

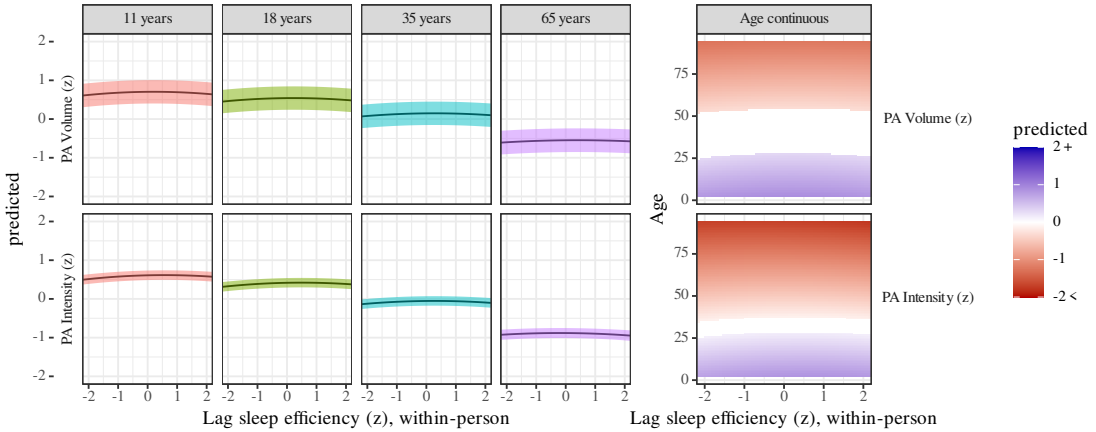


Figure 4: Standardised physical activity predicted by within-person sleep efficiency the previous night, with between-person sleep efficiency held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

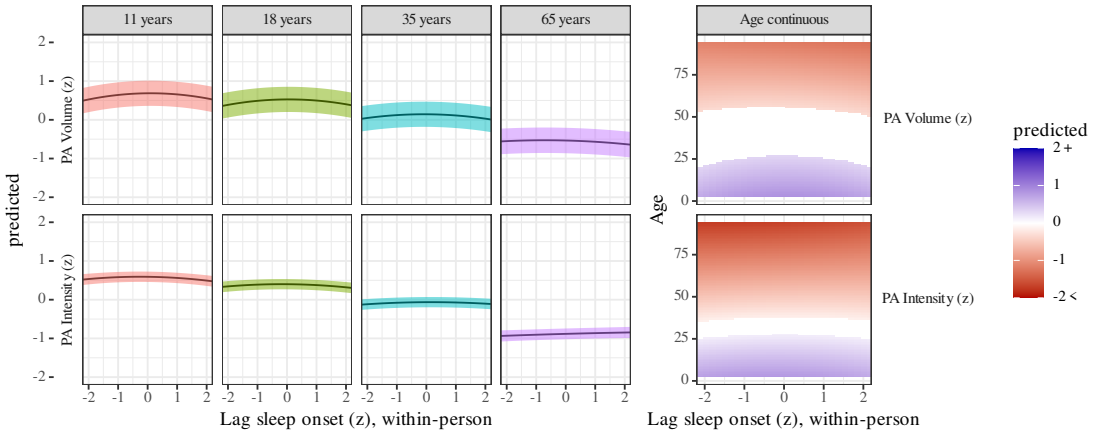


Figure 5: Standardised physical activity predicted by within-person sleep onset the previous night, with between-person sleep onset held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

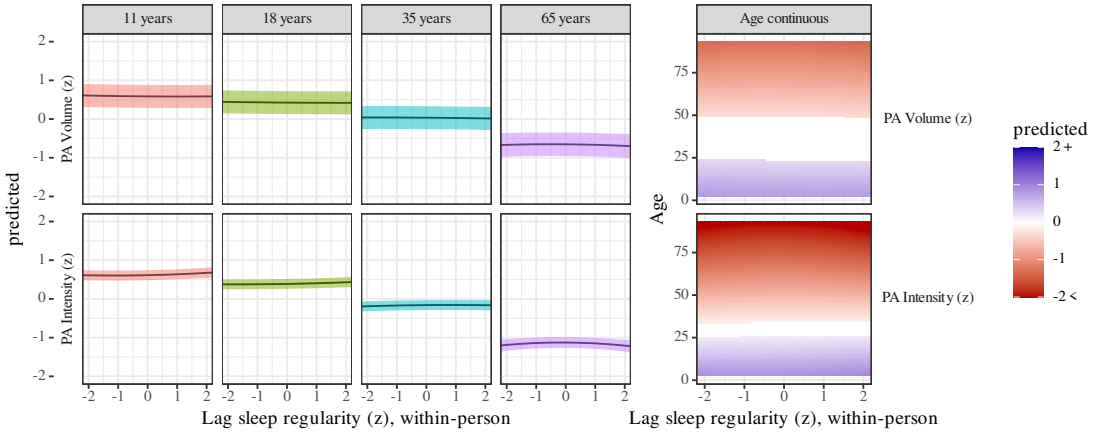


Figure 6: Standardised physical activity predicted by within-person sleep regularity over the two preceding calendar days, with between-person sleep regularity held at the sample mean. The panels on the left show the curvilinear relationship at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Discussion

Pooling accelerometer data from 20 studies spanning the lifespan, we found that on days when people were more active than usual, they slept longer and fell asleep earlier that night, from childhood to older adulthood. These associations were small. A 1 SD (23.7 mg) higher volume than usual, about twice a typical day-to-day fluctuation (SD 11.2 mg), was associated with 9.6 min longer sleep and 12 min earlier sleep onset at age 11, and with 8.6 min and 20 min, respectively, at age 65. In terms of MVPA, 30 minutes more than usual was associated with sleep onset 4–5 min later and a change in sleep duration of no more than 3.0 min. In the reverse direction, a night's sleep was barely associated with next-day activity.

Our within-person findings agree in direction with meta-analyses of experimental research showing that physical activity improves sleep, although the associations were much weaker.[7,8] For example, Kredlow and colleagues[7] found small ($0.22 - 0.30$) but significant relationships between physical activity and sleep duration and efficiency in adults using experimental studies for both acute and regular physical activity. These effect sizes are not directly comparable, however, as the experiments manipulated physical activity, whereas we examined natural day-to-day variation. Accelerometry also distinguishes quiet wakefulness from sleep less well than polysomnography, particularly in children,[22] which may attenuate associations, and the benefits of exercise for sleep are concentrated in people whose sleep is poor at baseline,[8] whom we did not target. That more MVPA than usual was associated with slightly later sleep onset, unlike overall volume, is consistent with evidence that strenuous exercise close to bedtime can delay sleep.[25]

Among observational studies of day-to-day sleep and physical activity, a meta-analysis of 33 studies in adults did not support a bidirectional daily association,[14] and a meta-analysis of 11 studies in children and youth found weak and inconsistent associations of physical activity with sleep.[15] Two recent studies of consumer wearables found that previous-day activity had little association with sleep,[16] that shorter sleep was followed by more activity the next day,[16,17] and that earlier-than-usual sleep onset was followed by more moderate-to-vigorous activity.[17] Our findings for MVPA match these previous studies: earlier and shorter sleep than usual were followed by slightly more MVPA the next day. Unlike the wearable studies, we found consistent associations of activity with that night's sleep, possibly reflecting differences in devices, sleep algorithms, and populations. Between-person associations were larger and changed more with age than within-person associations, as in very old adults,[26] which shows why the two need to be separated.

Because the regularity index spans two calendar days and counts daytime rest as sleep, an unusually active or inactive day changes it regardless of sleep, which explains why next-day activity predicted it as strongly as same-day activity and why both extremes predicted lower regularity. Once the index no longer overlapped the day being predicted, more regular sleep was not followed by more activity, and the between-person association may also partly reflect daytime rest. Because irregular sleep predicts mortality and cardiovascular events independently of sleep duration,[27,28] studies of physical activity and sleep regularity need measures that separate sleep from daytime rest.

Within-person associations with sleep duration and onset were present from childhood to older adulthood and varied little with age, whereas several between-person associations, including that with sleep duration, weakened or reversed with age. That the day-level associations were similar in children and adults suggests that the clearer evidence from meta-analyses in adults than in children and adolescents[6,7,10,11] may reflect study design or measurement rather than age.

Strengths of our study include device-measured sleep and physical activity, harmonised from raw data collected with 24-hour wear protocols across multiple studies, regions, and age groups, and the separation of within- from between-person associations. It also has limitations. Despite our efforts to harmonise data, variability in accelerometer wear locations and slight differences in data collection protocols across studies may have introduced measurement inconsistencies. Physical activity estimates from different wear locations are strongly correlated but agree less well in absolute terms, particularly between wrist and hip,[29] and MVPA cut-points differed by age group, device, and wear location. Restricting the sample to participants with at least seven

valid days gave similar within-person associations at age 65 but weaker ones at age 11, for both sleep duration (β = 0.07 [0.06, 0.09]) and onset (β = −0.04 [−0.05, −0.03]). Within-person estimates are free of stable differences between people but not of factors that vary from day to day, such as illness, school or work schedules, and weekends, so they do not establish causality. Adjusting for the previous day's outcome changed little, although such adjustment can be biased when participants contribute few days. Finally, adolescents and young adults were under-represented, which limits generalisability to these groups and matters because young adults are the most likely to report difficulty falling asleep.[4]

These findings have implications for public health recommendations. Guidelines do not recommend physical activity to improve sleep in children and adolescents, yet we found that people of all ages slept slightly longer and earlier after more active days. The gains were small, so physical activity is better presented as a consistent support for sleep than as a remedy for poor sleep. We did not examine these relationships specifically in individuals with sleep disorders or other health conditions, where the effects of physical activity may differ. Future research should test whether increasing physical activity lengthens sleep in children and adolescents, and should measure sleep regularity with methods that separate sleep from daytime rest.

Contributors

CL, TH, TS, BdPC, PP, and MN conceived and designed the study. TH managed the data acquisition. TS managed the data harmonisation and developed the accelerometer processing pipeline. JC, TS, and MN conducted the statistical analyses. TH, TS, JC, and MN had direct access to and verified the underlying data. TH, TS, BdPC, MN, PP, JC, and CL drafted the manuscript. All co-authors not individually listed, contributed data from their original studies. All authors contributed to the interpretation of results and critically revised the manuscript for important intellectual content. All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Declaration of interests

We declare no competing interests.

Data sharing

Individual participant data remains the property of the original study investigators, and we cannot share it without their permission. Reasonable requests for access to the pooled dataset will be considered by the corresponding author, and we will seek permission from the original investigators. Related documents, including the study protocol and analysis plan, are publicly available (https://osf.io/ahd96, https://github.com/Motivation-and-Behaviour/sleepIPD_analysis).

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AI declaration

We used AI tools, primarily Claude Opus 5.5, to assist with code review and editing of the analysis code. The authors had full oversight of the analysis and code, and all decisions were made by the authors.

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